Aniketh Girish | Privacy Researcher

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Research Summary

I'm a Postdoctoral Researcher at the IMDEA Networks Institute in Madrid, Spain. my research sits at the intersection of privacy, empirical network measurement, and platform governance, with a focus on uncovering hidden data flows and cross-platform tracking in modern smart ecosystems. I have exposed covert tracking vectors—from localhost-based web-to-app tracking used by Meta and Yandex to wireless-scanning SDKs that infer location and unauthorized local-network data flows in smart homes. My measurement-driven research has influenced platform practices (e.g., Android 16's local-network permission) and informed EU data protection authorities. It was also cited by Spain's Prime Minister in discussions that led to a Congressional inquiry into Meta under the Digital Services Act for their localhost-based tracking practices. I publish in top-tier venues including PETS, IMC, and USENIX Security. My work has also been covered internationally, including by The Washington Post, Wired, CBC News, El País, and 100+ other outlets.

Research Interests

- Security and privacy for emerging technologies (mobile/web/smart-home ecosystems) with emphasis on covert tracking and cross-platform data flows.
- Ads privacy, abuse and risk measurement; network security of connected consumer devices.
- Unifying technical measurement with technology policy and regulatory perspectives.
- Methods: on-device and in-network measurement, large-scale traffic analysis, cross-vantage experimentation, and empirical policy evaluation.

Professional Experience

IMDEA Networks Institute — **Madrid, Spain**
Postdoctoral Researcher — *10/2025–Present*
Advisor: Prof. Narseo Vallina-Rodriguez.

Khoury College, Northeastern University — **Boston, MA, USA**
Visiting Researcher — *04/2024 - 05/2024*
Advisor: Prof. David Choffnes.

Internet Initiative Japan — **Tokyo, Japan**
Research Associate — *06/2019 - 08/2019*
Advisor: Dr. Hajime Tazaki.

Rochester Institute of Technology — **Rochester, NY, USA**
Research Intern — *05/2018–05/2019*
Advisor: Prof. Taejoong (Tijay) Chung.

Education

IMDEA Networks Institute & UC3M — **Madrid, Spain**
Ph.D. Telematics Engineering — *2020–2025*
Dissertation: *Understanding Interconnected Abuse in Consumer Smart Device Ecosystems.* Graduated *Sobresaliente Cum Laude.*
Advisors: Dr. Srdjan Matic, Prof. Narseo Vallina-Rodriguez.

- Challenged the existing process-centric view of privacy controls, showing that violations often arise from systemic interactions across digital ecosystems rather than isolated components.
- Independently ideated and executed the research end-to-end, developing empirical, multi-vantage measurement approaches to uncover cross-device and cross-platform tracking pathways and unmonitored data flows.
- Drove real-world impact, leading to platform-level protections (e.g., Android 16's local-network permission), privacy fixes from Google, TP-Link, Philips, and 20+ IoT vendors, and attracting global attention with 100+ international news and media reports.

University of Carlos III Madrid (UC3M) — **Madrid, Spain**
M.Sc. Computer Security — *2020–2021*
Dissertation: *ImposTer: Towards an extensible privacy analysis framework for Smart Home ecosystem.*
Advisors: Dr. Srdjan Matic, Prof. Juan Tapiador, Prof. Narseo Vallina-Rodriguez.

Amrita Vishwa Vidyapeetham **Kerala, India**
B.Tech. Computer Science and Engineering *2016–2020*
Dissertation: *Empirical Analysis of Adaptive Bitrate Streaming over QUIC.*
Semester abroad: Ben-Gurion University of the Negev, Israel (Summer 2018).

Research Awards and Honors

2022: **Traffic Measurement and Analysis (TMA)** — Best poster and presentation award.

Selected Publications

Conference Proceedings

[1] T. Vlummens*, **A. Girish***, N. Weerasekara, F. Zuiderveen Borgesius, G. Acar, and N. Vallina Rodriguez, "Bridges to self: Silent web-to-app tracking on mobile via localhost," in *Proceedings of the USENIX Security Symposium*, 2026. [Online]. Available: https://localmess.github.io/.

[2] **A. Girish**, J. Reardon, J. Tapiador, S. Matic, and N. Vallina-Rodriguez, "Your signal, their data: An empirical privacy analysis of wireless-scanning sdks in android," in *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2025. [Online]. Available: https://petsymposium.org/popets/2025/popets-2025-0103.pdf.

[3] N. Alomar, J. Reardon, **A. Girish**, N. Vallina-Rodriguez, and S. Egelman, "The effect of platform policies on app privacy compliance: A study of child-directed apps," in *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2025. [Online]. Available: https://petsymposium.org/popets/2025/popets-2025-0094.pdf.

[4] **A. Girish***, T. Hu*, V. Prakash, D. Dubiois. J, S. Matic, D. Y. Huang, S. Egelman, J. Reardon, J. Tapiador, D. Choffnes, and N. Vallina-Rodriguez, "In the room where it happens: Characterizing local communication and threats in smart homes," in *Proceedings of the Internet Measurement Conference (IMC)*, 2023. DOI: 10.1145/3618257.3624830.

[5] **A. Girish**, V. Prakash, S. Egelman, J. Reardon, J. Tapiador, D. Y. Huang, S. Matic, and N. Vallina-Rodriguez, "Challenges in inferring privacy properties of smart devices: Towards scalable multi-vantage point testing methods," in *Proceedings of the International CoNEXT Student Workshop*, 2022. [Online]. Available: https://doi.org/10.1145/3565477.3569148.

[6] **A. Girish**, J. Tapiador, S. Matic, and N. Vallina-Rodriguez, "Towards an extensible privacy analysis framework for smart homes," in *Proceedings of the Internet Measurement Conference (IMC)*, 2022. [Online]. Available: https://doi.org/10.1145/3517745.3563023.

[7] H. Lee, **A. Girish**, R. van Rijswijk-Deij, T. T. Kwon, and T. Chung, "A longitudinal and comprehensive study of the DANE ecosystem in email," in *Proceedings of the USENIX Security Symposium*, 2020. [Online]. Available: https://www.usenix.org/conference/usenixsecurity20/presentation/lee-hyeonmin.

Journal Articles

[8] K. Hageman, A. Feal, J. Gamba, **A. Girish**, J. Bleier, M. Lindorfer, J. Tapiador, and N. Vallina-Rodriguez, "Mixed signals: Analyzing software attribution challenges in the android ecosystem," *IEEE Transactions on Software Engineering*, 2023. DOI: 10.1109/TSE.2023.3236582.

* Co-first authors.

Student Grants and Travel Awards

2023: **Summer Research Institute on Security and Privacy (EPFL SURI)** — Lausanne, Switzerland — Student Travel Grant.

2023: **Interdisciplinary Summerschool on Privacy (ISP)** — Nijmegen, The Netherlands — Student Travel Grant.

2022: **Internet Measurement Conference (IMC)** — Nice, France — Student Travel Grant.

2022: **Traffic Measurement and Analysis (TMA)** — Enschede, The Netherlands — Student Travel Grant.

2018: **BGU Summer School Scholarship** — Ben-Gurion University of the Negev, Israel.

Invited Talks

2025: **EURECOM Mobsec Seminar** — Covert Web-to-App Tracking via Localhost on Android. (Sophia Antipolis, Remote)

2025: **Ad-Filtering Dev Summit** — Covert Web-to-App Tracking via Localhost on Android. (Limmassol, Cyprus)

2025: **IETF 123 (PEARG)** — Covert Web-to-App Tracking via Localhost on Android. (Madrid, Spain)

2025: **XIX RediMadrid Conference** — Your Signal, Their Data: An Empirical Privacy Analysis of Wireless-scanning SDKs in Android. (Madrid, Spain)

2024: **RIPE 89 Plenary** — In the Room Where It Happens – Characterizing Local Communication and Threats in Smart Homes. (Prague, Czech Republic)

2024: **XIX RediMadrid Conference** — In the Room Where It Happens – Characterizing Local Communication and Threats in Smart Homes. (Madrid, Spain)

2020: **BSides India** — Email security with DANE: Empirical Analysis of DANE for SMTP in the wild. (Delhi, India)

2019: **IIESoc RFCsWeLove** — DOH? DONUT! - The new internet according to RFC 8484. (Bangalore, India)

2019: **FOSSDEM'19** — DOH? DONUT! - The new internet according to RFC 8484. (Brussels, Belgium)

2019: **Akademy'19** — Strengthen Code Review Culture: rm -rf 'Toxic Behaviors'. (Milan, Italy)

2018: **Akademy'18** — Strengthen Code Review Culture: rm -rf 'Toxic Behaviors'. (Vienna, Austria)

2017: **conf.KDE.in '17** — Object tracking using OpenCV and Qt. (Guwahati, India)

Research Projects

2021–2024: **TRUST aWARE (EU H2020, €4.6M, Award #101021377)** — Researcher. Enhancing digital security, privacy, and trust in software.

Open Source Contributions

2019: **Google Code-In | KDE** — Org Admin mentoring pre-university contributors across KDE projects.

2018: **Google Summer of Code | GNU Wget2** — Implemented DNS-over-HTTPS (DoH) with IPv4/IPv6, EDNS, and DNS-SD support.

2017: **Google Summer of Code | KDE/Krita** — Built encrypted client-server data sharing to prevent spam and secure transfers.

Advisory Roles

2022–2023: **IoXt Privacy Compliance Program** — Contributed strategy and testing frameworks for a global IoT security compliance program.

2016–2020: **FOSS Club Mentor, Amrita Vishwa Vidyapeetham** — Mentored students on open-source development and ran workshops.

Selected Media Coverage

Localmess Attack.....

06/2025: **ArsTechnica** — Meta and Yandex are de-anonymizing Android users' web browsing identifiers.

06/2025: **Washington Post** — Meta found a new way to violate your privacy. Here's what you can do.

06/2025: **El País** — Este es el método oculto con el que Meta rastrea sin permiso la navegación en móviles (también en modo incógnito o con VPN). (Spanish)

06/2025: **Sky News** — Meta found 'covertly tracking' Android users through Instagram and Facebook.

06/2025: **ABC** — Cómo Meta y Yandex consiguen datos concretos sobre tus hábitos en internet sin consentimiento. (Spanish)

06/2025: **De Morgen** — Het sluwe achterpoortje waarmee u online gevolgd wordt: Meta deed er alles aan om dit verborgen te houden. (Dutch)

06/2025: **Schneier on Security** — New Way to Covertly Track Android Users.

06/2025: **Zeroparty Data** — "Localhost tracking" explained. It could cost Meta 32 billion.

06/2025: **Security Boulevard** — Meta's Secret Spyware: 'Local Mess' Hack Tracks You Across the Web.

06/2025: **Deutschland Funk** — App-Attacke - Europäische Forscher decken groben Privacy-Verstoß von Meta auf. (German)

06/2025: **The Register** — Meta pauses mobile port tracking tech on Android after researchers cry foul.

06/2025: **TechRepublic** — Meta, Yandex Covertly Tracked Billions of Android Users, According to Researcher.

Wireless Beacons.....

07/2025: **El País** — How does your phone know so many private details? How thousands of apps are exploiting mysterious indoor location tracking.

07/2025: **La Sexta Noticias** — ¿Alguna vez has pensado que el móvil te espía? Confirmado, apps te siguen usando WIFI, Bluetooth y datos aunque no actives el GPS .

IoT Security and Privacy.....

11/2023: **ABC** — Espías en casa: la cara oscura de internet de las cosas en los 'hogares inteligentes'. (Spanish)

11/2023: **elcorreo** - Los dispositivos que tenemos en casa hablan entre sí y no te va a gustar lo que se cuentan. (Spanish)

11/2023: **HiddenWires** — New research reveals privacy and security threats in smart homes.

11/2023: **La Nueva Espana** - Así nos vigilan los dispositivos inteligentes en casa: saben dónde vivimos, qué hacemos en cada momento y hasta pueden averiguar nuestro nivel de renta. (Spanish)

10/2023: **CBS News Calgary** - Are your home's smart devices leaking private data? Calgary researcher set to find out.

10/2023: **Wired** - Los dispositivos de tu hogar inteligente te espían, advierte este informe. (Spanish)

10/2023: **El País** — 'People have no idea': How smart devices spy on us and reveal information about our homes.